



EEE4022S

Final year project topics

2025

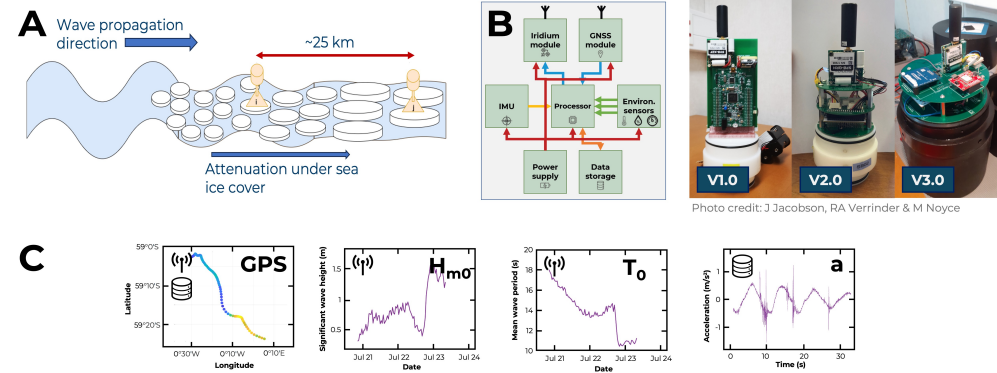
RA Verrinder

Topic list

- **RAV2025-01** SHARC project: Integration of GPS wave measurement into ice-tethered buoys
- **RAV2025-02** SHARC project: Sea ice surface roughness mapping with LiDAR and super-resolution imaging
- **RAV2025-03** SHARC project: Acoustic-visual-inertial odometry for under-ice navigation
- **RAV2025-04** SHARC project: Robust power management design [SELF PROPOSED]
- **RAV2025-05** Adaptive path planning for Wave Gliders: AIS-based dynamic collision avoidance
- **RAV2025-06** Automated quantitative analysis of mycobacterial cording phenotypes
- **RAV2025-07** Remote weight monitoring system for top hives in the Zambian Miombo woodlands

Note

If you are interested in selecting one of these topics, please [message me on MS Teams](#) if you need more information and use my [booking page](#) to book a meeting to discuss the project details.

Student Proposed?	Y/N	N/A
ID	RAV2025-01	
Supervisor	RA Verrinder	
Co-Supervisor(s)	-	
Title	SHARC project: Integration of GPS wave measurement into ice-tethered buoys	
Description	Background Sea ice observation data are crucial to understanding the extent of climate perturbations and to inform global models that predict future climate and ocean dynamics. Sea ice variability and extent are largely determined by the marginal ice zones (MIZ), areas of transitional sea ice found between the open ocean and solidified pack ice. These regions are generated through the seasonal formation and subsequent melting of various sea ice types and include distinct stages of ice consolidation such as grease, frazil, pancake, and pack ice, each having specific properties (such as concentration and thickness) and dynamics (such as inter-ice, atmosphere-ice-ocean interactions and drift) (Vichi, 2022). The heterogeneity in sea ice types and behaviours, as well as the remote and harsh environment, make the MIZ a complex system to capture observationally. Within this challenge lies an opportunity to improve the in situ observation and characterisation of sea ice properties through the enhancement of autonomous sea-ice-tethered instruments. A variety of interacting forces, including local wind and wave conditions, strongly affect the formation and development of sea ice in the MIZ. To better characterise sea ice development under the influence of wave dynamics and turbulent environmental conditions, in situ measurements and observations are needed. Wave-in-ice measurement is essential to understand the interaction between ocean waves and sea ice, which influences sea ice breakup and melting processes. Wave attenuation in sea ice is influenced by factors such as ice flexure, ice thickness, turbulence, and wave scattering. Recent studies have shown that wave attenuation is dominated by dissipation rather than scattering by ice floes. This dissipation is highly variable and related to the size, thickness, and consolidation of ice floes. Understanding these processes is critical for understanding sea ice variability and predicting seasonal sea ice extent (Raghukumar et al., 2019; Hošeková et al., 2020) and accurate wave measurement is critical to characterise wave attenuation through the sea ice field (Thomson et al., 2021; Ardhuin et al., 2020). Waves in the ocean are traditionally measured using buoys with onboard motion sensors such as accelerometers and gyroscopes. These sensors detect the buoy's accelerations and angular velocities, which are then integrated to derive wave parameters like significant wave height, mean wave period, and wave direction. However, this method requires precise calibration and can be affected by noise and drift over time (Thomson et al., 2021). In contrast, GPS-based wave measurement uses the Global Positioning System to track the precise movements of a buoy in three dimensions. This method improves the accuracy of wave measurements, particularly for horizontal displacements, and reduces the need for frequent calibration and complex data processing (Herbers et al., 2012).  Fig. 1. Overview of waves-in-ice measurement using the UCT SHARC buoy. (A) Wave attenuation through sea ice is influenced by sea ice concentration, thickness, and type. UCT SHARC buoys were deployed at 25 km intervals in the MIZ in the SCALE 2022 Winter Cruise to characterise wave attenuation in the Antarctic Marginal Ice Zone. (B) The UCT SHARC buoy V3.0 includes onboard GPS, inertial, and environmental sensors, with onboard data storage and processing capabilities as well as Iridium satellite communication. (C) The raw inertial time-series and GPS-based drift measurements are stored onboard the device, where they are used to calculate significant wave height (H_{m0}) and mean wave period (T_0), which are transmitted at regular intervals using Iridium. (Image credit: MP Noyce and RA Verrinder) Subject of investigation The Department of Electrical Engineering (UCT) and the Department of Oceanography (UCT) have developed an ice-tethered instrument (SHARC - Southern Hemisphere Antarctic Research Collective buoy) (Jacobson, 2024; Noyce, 2024), which can be deployed on ice floes in the MIZ of the Southern Ocean to measure ice drift, wave-in-ice motion, and environmental effects. Our platform uses motion-based wave measurement to determine significant wave height and mean wave period (Noyce et al., 2023; Noyce, 2024). To enhance the capabilities of the existing motion-based SHARC buoy system, this project aims to integrate GPS wave measurement techniques (Herbers et al., 2012). By combining GPS with the current accelerometer-based system, the enhanced SHARC buoys will be able to accurately capture both horizontal and vertical wave displacements.	

Deliverables	The key objectives of this project are: (a) To understand the full scope and requirements of the project. (b) To conduct a literature review of previously published research in this field and critically evaluate current technologies and research that applies to this project. (c) To investigate GPS-based directional wave measurement techniques, and select the most appropriate approach for a power-constrained embedded platform i.e. the SHARC buoy. (d) To develop a test platform to evaluate the proposed approach. (e) To develop and implement data processing algorithms for the new integrated system. (f) To test and evaluate the system performance and compare it to previously implemented solutions. (g) To discuss the performance of the system, draw conclusions, and make recommendations for future improvements. The key deliverables of this project are: (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) A fully integrated test bed with GPS-based wave measurement capability. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (f) A poster summarising the research project report. (g) A seminar to the Department summarising the research project report. (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Signal processing, Interest in oceanography
Extra Information	References F. Ardhuin, M. Otero, S. Merrifield, A. Grouazel, and E. Terrill. Ice break-up controls dissipation of wind-waves across Southern Ocean sea ice. <i>Geophysical Research Letters</i>, 47:e2020GL087699, Apr 2020. doi: 10.1029/2020GL087699 T. H. C. Herbers, P. F. Jessen, T. T. Janssen, D. B. Colbert, and J. H. MacMahan. Observing ocean surface waves with GPS-tracked buoys. <i>Journal of Atmospheric and Oceanic Technology</i>, 29(7):944–959, Jul 2012. doi: 10.1175/JTECH-D-11-00128.1 L. Hošeková, M. P. Malila, W. E. Rogers, L. A. Roach, E. Eidam, and L. Rainville. Attenuation of ocean surface waves in pancake and frazil sea ice along the coast of the chukchi sea. <i>Journal of Geophysical Research: Oceans</i>, 125(12): e2020JC016746, Dec 2020. doi: 10.1029/2020JC016746 J. Jacobson. SHARC Buoy: Robust firmware design for a novel, low-cost autonomous platform for the Antarctic Marginal Ice Zone in the Southern Ocean. MSc thesis, Marine and Antarctic Research Centre for Innovation and Sustainability (MARiS), University of Cape Town, 2024. url: http://hdl.handle.net/11427/35788 M. Noyce, R. Verrinder, and M. Vichi. Identification of ice floe impacts in an inertial time series and implications for wave measurement in pancake ice. In <i>IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium</i>, pages 149–152. IEEE, 2023. doi: 10.1109/IGARSS52108.2023.10281906 M. Noyce. UCT SHARC Buoy V3.0: Waves-in-ice measurement in the Antarctic MIZ. MSc thesis, Marine and Antarctic Research Centre for Innovation and Sustainability (MARiS), University of Cape Town, 2024. url: http://hdl.handle.net/11427/40386 K. Raghukumar, G. Chang, F. Spada, C. Jones, T. Janssen, and A. Gans. Performance characteristics of “Spotter,” a newly developed real-time wave measurement buoy. <i>Journal of Atmospheric and Oceanic Technology</i>, 36(6): 1127–1141, 2019. doi: 10.1175/JTECH-D-18-0151.1 J. Thomson, L. Hošeková, M. H. Meylan, A. L. Kohout, and N. Kumar. Spurious rollover of wave attenuation rates in sea ice caused by noise in field measurements. <i>Journal of Geophysical Research: Oceans</i>, 126:e2020JC016606, Mar 2021. doi: 10.1029/2020JC016606 M. Vichi. An indicator of sea ice variability for the Antarctic marginal ice zone. <i>The Cryosphere</i>, 16:4087–4106, 2022. doi: 10.5194/tc-16-4087-2022
Broad Research Area	Sea ice observation, Wave measurement, Embedded Systems, GPS
Project suitable for: ME/ECE/EE/ALL?	ME, ECE

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EEE4022S Topic RAV2025-01 GAs

Student Proposed?	Y/N	N/A
ID	RAV2025-01	
Supervisor	RA Verrinder	
Co-Supervisor(s)	-	
Title	SHARC project: Integration of GPS wave measurement into ice-tethered buoys	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the integration of GPS-based wave measurement into the existing SHARC buoy system. It involves ill-posed and under-specified problems, such as the: - refinement of GPS integration into the SHARC buoy, - development of GPS wave measurement data processing algorithms, which can be run on an embedded system. It also presents high-level problems, such as: - Understanding wave measurement in the Antarctic marginal ice zone (MIZ). - Calibrating sensors and ensuring accurate wave measurement in varying sea ice conditions. - Handling harsh environmental factors and power constraints - Solutions for integrating GPS technology into the SHARC buoy system, which is not trivial. The project also necessitates handling complex and incomplete information by: - Synthesising information from various sources such as oceanographic data and sensor specifications. - Balancing technical challenges, engineering design, and stakeholder interests.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: - Design and integrate GPS-based wave measurement into the SHARC buoy system. - Analyse data collected from GPS and motion sensors to validate the wave measurement algorithms - Compare the performance of the integrated system with existing solutions. - Utilise statistical analysis to interpret data and validate the measurement techniques.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be computational in nature and will require competencies in data handling, sensor integration, data analysis, and embedded programming. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate data handling processes and tools to achieve project objectives. Software required may include: - C for embedded programming of the firmware - MATLAB/Python for data processing and analysis - MATLAB/Python for modelling and simulation - MATLAB/Python for data validation	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2025-01

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined here .		X
Answer is No here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer Yes to any of the above questions.			
If the answer is Yes to Q1 , please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is Yes to Q2 , please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to Q9 is Yes , do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is Yes to Q5 , please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If Yes , provide details of the issue and your plan to solve it]		X

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Student Proposed?	Y/N	N/A
ID	RAV2025-02	
Supervisor	RA Verrinder	
Co-Supervisor(s)	-	
Title	SHARC project: Sea ice surface roughness mapping with LiDAR and super-resolution imaging	
Description	Background The distribution and behaviour of sea ice within the Marginal Ice Zone (MIZ) of the Southern Ocean significantly affect global climate processes. Despite its importance, our understanding of the sea ice life cycle in this dynamic and transitional region remains constrained by the limited availability of in situ observational data, particularly during the harsh Antarctic winter. Sea ice within the MIZ is subject to continuous modification due to interactions with local wind, ocean currents, and wave activity, resulting in a wide range of surface morphologies (Vichi, 2022). One critical but under-characterised parameter in this context is surface roughness, which influences the momentum exchange between the atmosphere, ice, and ocean. These exchanges impact ice drift, deformation, and melt processes, making surface roughness a key variable in sea ice dynamic modelling. Additionally, roughness modulates the back-scatter signal received by satellite-borne sensors, affecting the accuracy of remote sensing estimates of sea ice concentration and type (Landy et al., 2015, 2023). Traditional methods for estimating sea ice roughness have been limited in spatial resolution and coverage. However, recent studies, such as by Landy et al. (2015, 2023), demonstrate that terrestrial LiDAR systems can be used to capture centimetre-scale surface topography across various sea ice types. Complementing this, advances in camera-based super-resolution texture reconstruction techniques now allow for the generation of highly detailed surface representations from low-resolution imagery. The fusion of LiDAR point clouds with super-resolved image textures presents a promising approach for reconstructing both the geometric and visual surface properties of sea ice floes. A 3D imaging process B 3D model	

Deliverables	The key objectives of this project are: (a) To understand the full scope and requirements of the project. (b) To conduct a literature review of previously published research in this field and critically evaluate current techniques and research that applies to this project. (c) To investigate surface roughness measurement techniques, and select the most appropriate approach for this application. (d) To develop a experimental platform to collect data for the proposed approach. (e) To develop and implement data processing algorithms. (f) To test and evaluate the system performance and compare it to previously implemented solutions. (g) To discuss the performance of the system, draw conclusions, and make recommendations for future improvements. The key deliverables of this project are: (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) An investigation into surface roughness extraction techniques using LiDAR and camera data. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (f) A poster summarising the research project report. (g) A seminar to the Department summarising the research project report. (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Signal processing, Interest in marine science
Extra Information	References D. Jones. SHARC project: Ice floe size and shape characterisation from LiDAR scans. BSc(Eng) final year project, Department of Electrical Engineering), University of Cape Town, 2024. url: https://tinyurl.com/djones2023 J. Landy, A. Komarov, and D. Barber. Numerical and experimental evaluation of terrestrial LiDAR for parameterizing centimeter-scale sea ice surface roughness. <i>IEEE Transactions on Geoscience and Remote Sensing</i>, 53: 4887–4898, 2015. doi: 10.1109/TGRS.2015.2412034 J. Landy, A. Komarov, and D. Barber. Secondary-scale surface roughness parameterization using terrestrial LiDAR. In <i>IGARSS 2015-2015 IEEE International Geoscience and Remote Sensing Symposium</i>, pages 2068–2071. IEEE, 2023. doi: 10.1109/igarss.2015.7326208 A. Spirakis. Investigation into surface roughness characterisation using LiDAR and cameras. BSc(Eng) final year project, Department of Electrical Engineering), University of Cape Town, 2021. url: https://tinyurl.com/spirakis2021 A. Spirakis, J. Hepworth, and R. Verrinder. 3D reconstruction of pancake sea ice using LiDAR and cameras. In <i>IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium</i>, pages 60–63. IEEE, 2023. doi: 10.1109/igarss52108.2023.10282250 M. Vichi. An indicator of sea ice variability for the Antarctic marginal ice zone. <i>The Cryosphere</i>, 16:4087–4106, 2022. doi: 10.5194/tc-16-4087-2022
Broad Research Area	Sea ice observation, image processing, 3D reconstruction
Project suitable for: ME/ECE/EE/ALL?	ECE

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EEE4022S Topic RAV2025-02 GAs

Student Proposed?	Y/N	N/A
ID	RAV2025-02	
Supervisor	RA Verrinder	
Co-Supervisor(s)	-	
Title	SHARC project: Sea ice surface roughness mapping with LiDAR and super-resolution imaging	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the investigation of surface roughness characterisation using LiDAR and camera images. It involves ill-posed and under-specified problems, such as the: • the fusion of LiDAR and camera images, • development of super-resolution algorithms, which are applicable for this application. It also presents high-level problems, such as: • Understanding surface roughness measurement. • Calibrating sensors and ensuring accurate sensor fusion measurement in varying sea ice conditions. The project also necessitates handling complex and incomplete information by: • Balancing technical challenges, engineering design, and stakeholder interests.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: • Design surface roughness extraction pipeline. • Analyse data collected from sensors to validate the algorithms • Compare the performance of the integrated system with existing solutions. • Utilise statistical analysis to interpret data and validate the measurement techniques.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be computational in nature and will require competencies in data handling, sensor integration and data analysis. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate data handling processes and tools to achieve project objectives. Software required may include: • MATLAB/Python for data processing and analysis • MATLAB/Python for modelling and simulation • MATLAB/Python for data validation	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2025-02

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?	X	
Q4	Does it exceed the minimum risk defined here .		X
Answer is No here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer Yes to any of the above questions.			
If the answer is Yes to Q1 , please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is Yes to Q2 , please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to Q9 is Yes , do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is Yes to Q5 , please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If Yes , provide details of the issue and your plan to solve it]		X

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EEE4022S RAV2025-03

Student Proposed?	Y/N N/A
ID	RAV2025-03
Supervisor	RA Verrinder
Co-Supervisor(s)	-
Title	SHARC project: Acoustic-visual-inertial odometry for under-ice navigation
Description	Background Autonomous and remotely operated platforms are increasingly being employed to extend our observational capabilities in harsh and inaccessible environments. In polar regions, under-ice exploration remains one of the most challenging and under-instrumented areas in marine science. Optical imaging beneath sea ice offers valuable insights into ice morphology, basal roughness, algae distribution, and ice-ocean interface processes. However, under-ice operations are constrained by limited GPS access, complex light conditions, and unstable navigation environments (Zhao et al., 2022) The BlueROV2 (Blue Robotics, 2025), is a cost-effective research-class underwater Remotely Operated Vehicle (ROV) and has gained popularity in research communities due to its customisability, affordability, and ease of integration with external sensors. The UCT BlueROV2 includes the following sensor suite: a high quality stereo camera pair, a doppler velocity log (DVL), underwater illumination, an inertial measurement unit (IMU), depth sensor and side scan sonar allowing for advanced underwater image dataset collection and spatial intelligence research and development, a key focus area in the African Robotics Unit and the Marine and Antarctic Research centre for Innovation and Sustainability (MARIS). It has six T200 Thrusters, four arranged in a vector configuration and two in a vertical configuration, providing feedback stabilised, full six-degree-of-freedom motion underwater. While most commonly used for general underwater exploration, its potential as a testbed for advanced navigation and perception systems, particularly acoustic-visual-inertial (AVI) SLAM (Xu et al., 2025), remains under-explored in polar environments. AVI odometry combines data from onboard cameras, doppler velocity log (DVL) and inertial measurement units (IMUs) to estimate the vehicle's trajectory and orientation in GPS-denied conditions. This is particularly suitable for under-ice scenarios where external acoustic positioning systems (like LBL or USBL) are not viable.  Fig. 3. The Blue Robotics BlueROV2 Underwater Remotely Operated Vehicle (ROV). The configuration above shows the 6 thrusters, high power illumination modules and stereo cameras located in the inspection enclosure. Image taken from (Blue Robotics, 2025) Subject of investigation This project will pilot an acoustic-visual-inertial odometry system (Xu et al., 2025) on the UCT BlueROV2 platform in a controlled pool environment to evaluate its potential for application in under-ice navigation and imaging, as a first step towards developing robust navigation and imaging tools for under-ice applications. The focus will be on integrating and evaluating an onboard AVIO system that can process monocular video streams alongside DVL and inertial measurements to estimate the ROV's position.

Deliverables	The key objectives of this project are: (a) To understand the principles and challenges of acoustic-visual-inertial odometry (AVIO) in underwater GPS-denied environments. (b) To conduct a literature review of current AVIO techniques and evaluate their suitability for use on compact marine platforms such as the BlueROV2. (c) To implement a fused AVIO pipeline that combines inertial measurements, visual feature tracking, and acoustic positioning data. (d) To calibrate and optimise time synchronisation and spatial alignment between all sensing modalities. (e) To conduct controlled experiments in a pool environment to evaluate localisation drift, robustness, and accuracy under different motion profiles and visual/acoustic conditions. (f) To critically analyse the system performance, draw conclusions about feasibility for under-ice deployments, and make recommendations. The key deliverables of this project are: (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) A fully integrated BlueROV2-based AVIO system tested in a controlled pool environment. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (f) A poster summarising the research project report. (g) A seminar to the Department summarising the research project report. (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Image processing, Robotics, Interest in marine science
Extra Information	References Blue Robotics. BlueROV2. [ONLINE], 2025. Accessed: 2025-07-11. url: https://bluerobotics.com/store/rov/bluerov2/ S. Xu, K. Zhang, and S. Wang. AQUA-SLAM: Tightly coupled underwater acoustic-visual-inertial SLAM with sensor calibration. <i>IEEE Transactions on Robotics</i>, 41:2785–2803, 2025. doi: 10.1109/tro.2025.3554396 L. Zhao, M. Zhou, and B. Loose. Towards under-ice sensing using a portable ROV. In <i>OCEANS 2022-Hampton Roads</i>, pages 1–8. IEEE, 2022. doi: 10.1109/OCEANS47191.2022.9977140
Broad Research Area	Underwater robotics, SLAM
Project suitable for: ME/ECE/EE/ALL?	ME, ECE

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EEE4022S Topic RAV2025-03 GAs

Student Proposed?	Y/N	N/A
ID	RAV2025-03	
Supervisor	RA Verrinder	
Co-Supervisor(s)	-	
Title	SHARC project: Acoustic-visual-inertial odometry for under-ice navigation	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the integration of acoustic, visual, and inertial sensing for underwater odometry on the BlueROV2 platform. It involves ill-posed and under-specified problems, such as the: - development of sensor fusion software for underwater localisation. It also presents high-level problems, such as: - Maintaining localisation accuracy in underwater environments with limited visual features. - Calibrating multi-sensor systems for time and spatial alignment. - Handling noise and drift in inertial sensors and intermittent acoustic data. - Deploying a testable system in controlled aquatic environments as a proxy for under-ice conditions. The project also necessitates handling complex and incomplete information by: - Synthesising information from SLAM literature, underwater robotics research, and sensor specifications. - Balancing trade-offs between accuracy, computational load, and real-time feasibility.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: - Design and integrate an AVIO system into the BlueROV2 platform. - Develop and refine sensor fusion algorithms combining camera, IMU, and acoustic data. - Implement calibration routines for multi-sensor alignment. - Compare AVIO performance with single-modality approaches under varied test conditions. - Apply statistical analysis to evaluate localisation accuracy and drift across trials.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be computational and experimental in nature and will require competencies in sensor data acquisition, fusion algorithm development, embedded programming, and data analysis. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate signal processing and data integration tools to achieve project objectives. Software required may include: - C++ and ROS for interfacing with BlueROV2 and onboard sensors. - Python/MATLAB for data processing, calibration, and analysis. - OpenCV or similar libraries for visual tracking and feature extraction. - SLAM or state estimation frameworks (e.g., ORB-SLAM3) for odometry validation.	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation** and **experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

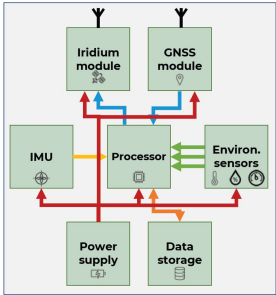
- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2025-03

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined here .		X
Answer is No here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer Yes to any of the above questions.			
If the answer is Yes to Q1 , please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is Yes to Q2 , please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to Q9 is Yes , do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is Yes to Q5 , please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If Yes , provide details of the issue and your plan to solve it]		X

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EEE4022S RAV2025-04

Student Proposed?	Y/N Y - GSBAC001
ID	RAV2025-04
Supervisor	RA Verrinder
Co-Supervisor(s)	-
Title	SHARC project: Robust power management design [SELF PROPOSED]
Description	Background The Southern Ocean Marginal Ice Zone (MIZ) remains one of the most dynamic and under-observed regions influencing global ocean-atmosphere interactions. In this zone, rapidly evolving sea ice conditions driven by wind, waves, and temperature variations make continuous monitoring particularly challenging—especially during the harsh austral winter (Vichi, 2022). Autonomous observing systems, such as ice-tethered buoys, have emerged as vital tools in bridging this data gap. These platforms offer invaluable in situ measurements that support sea ice modelling, climate analysis, and operational forecasting (Jacobson, 2024; Noyce et al., 2023; Noyce, 2024). Since 2019, the University of Cape Town (UCT) has developed and deployed several iterations of the SHARC (Southern Hemisphere Antarctic Research Collective) buoy system to track ice drift, wave-ice interactions, and environmental variability. SHARC ice-tethered instruments have been successfully deployed during multiple research campaigns, including aboard the SA Agulhas II and RV Polarstern (Jacobson, 2024; Noyce, 2024). Powering autonomous systems in the MIZ poses a significant engineering challenge, as batteries must operate reliably for extended durations in sub-zero temperatures without the possibility of recharging or maintenance, all while meeting the intermittent high-power demands of sensors and satellite communication modules. As the complexity of onboard instrumentation increases with the inclusion of imaging systems, environmental sensors, and telemetry, the design of a robust, efficient power management system becomes critical to ensure uninterrupted, long-term autonomous operation under severe environmental conditions (National Academies of Sciences, Engineering, and Medicine, 2022).   Photo credit: J Jacobson, RA Verrinder & M Noyce Fig. 4. Overview of the UCT SHARC buoy. The UCT SHARC buoy V3.0 includes onboard GPS, inertial, and environmental sensors, with onboard data storage and processing capabilities as well as Iridium satellite communication. Subject of investigation This project builds upon previous SHARC buoy prototypes (Jacobson, 2024; Noyce, 2024) and focuses on the design, simulation, and evaluation of a robust power management system to support long-term autonomous deployments in the Southern Ocean. The system must efficiently supply variable loads from sensors, telemetry modules, and embedded controllers, while operating from a non-rechargeable power source for mission durations of up to six months under extreme polar conditions, including sub-zero temperatures and fluctuating power demands. The power management system must implement real-time battery health monitoring, intelligent power scheduling, and switching between operational modes (e.g., acquisition, sleep, transmission). It must also be spatially compact, environmentally sealed, and highly reliable. The student will construct a detailed power budget, simulate the system's response to varying loads and temperatures, and prototype the design in a controlled laboratory setting. By the end of the project, a verified, power management system will have been tested, contributing to the next generation of SHARC buoy deployments with improved endurance and resilience.

Deliverables	The key objectives of this project are: (a) To understand the full scope and requirements of the SHARC buoy power system. (b) To conduct a literature review of existing polar-deployed power systems and current low-power embedded design techniques. (c) To construct a comprehensive power budget based on realistic mission profiles and operational requirements. (d) To design and simulate a robust, non-rechargeable power management system capable of sustaining multi-month Antarctic deployments. (e) To prototype and test the system under controlled laboratory conditions, including temperature variation, load switching, and fault injection scenarios. (f) To evaluate the performance of the system in terms of reliability, efficiency, thermal response, and failure modes. (g) To discuss the performance of the system, draw conclusions, and make recommendations for future improvements. The key deliverables of this project are: (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) A fully functional power management prototype capable of supporting a SHARC buoy mission scenario with data acquisition, storage, and telemetry loads. (d) A comprehensive power budget spreadsheet with mode-specific calculations and thermal/environmental considerations. (e) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (f) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (g) A poster summarising the research project report. (h) A seminar to the Department summarising the research project report. (i) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Power electronics, Embedded systems
Extra Information	References National Academies of Sciences, Engineering, and Medicine. <i>Technology developments to advance Antarctic research: Proceedings of a workshop</i>. The National Academies Press, 2022. doi: 10.17226/26699 J. Jacobson. SHARC Buoy: Robust firmware design for a novel, low-cost autonomous platform for the Antarctic Marginal Ice Zone in the Southern Ocean. MSc thesis, Marine and Antarctic Research Centre for Innovation and Sustainability (MARIS), University of Cape Town, 2024. url: http://hdl.handle.net/11427/35788 M. Noyce. UCT SHARC Buoy V3.0: Waves-in-ice measurement in the Antarctic MIZ. MSc thesis, Marine and Antarctic Research Centre for Innovation and Sustainability (MARIS), University of Cape Town, 2024. url: http://hdl.handle.net/11427/40386 M. Vichi. An indicator of sea ice variability for the Antarctic marginal ice zone. <i>The Cryosphere</i>, 16:4087–4106, 2022. doi: 10.5194/tc-16-4087-2022
Broad Research Area	Embedded Systems, Power Management, Interest in marine science
Project suitable for: ME/ECE/EE/ALL?	All

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EEE4022S Topic RAV2025-04 GAs

Student Proposed?	Y/N	N/A
ID	RAV2025-04	
Supervisor	RA Verrinder	
Co-Supervisor(s)	-	
Title	SHARC project: Robust power management design [SELF PROPOSED]	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the design and evaluation of a robust power management system for a polar-deployed autonomous buoy. It involves ill-posed and under-specified problems, such as: - Developing a compact, low-power, and efficient energy management strategy for unpredictable environmental conditions. - Accommodating diverse and time-varying power demands from multiple subsystems. It also presents high-level problems, such as: - Ensuring long-duration operation under sub-zero temperatures and extreme isolation with no user intervention. - Balancing reliability, energy efficiency, and size constraints for remote deployment. The project also necessitates handling complex and incomplete information by: - Synthesising hardware limitations, power usage data, and mission requirements into a coherent system design. - Considering thermal characteristics, system redundancy, and operational contingencies in a multidisciplinary context.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of engineering solutions to: - Develop a detailed power budget and operational profile. - Simulate and refine power delivery under various load and temperature conditions. - Prototype intelligent switching and scheduling mechanisms for power optimisation. - Integrate and validate battery health monitoring and fault detection features. - Evaluate system performance in a lab environment and compare against design specifications.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will involve significant computational and practical design elements, requiring skills in embedded system development, circuit simulation, and data logging. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting and applying appropriate design methodologies to achieve project objectives. Software and tools required may include: - Altium Designer/KiCad for PCB layout and power circuit design. - LTspice/Simulink for power system simulation. - C for embedded firmware development. - Python/MATLAB for data analysis, plotting, and power performance assessment.	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

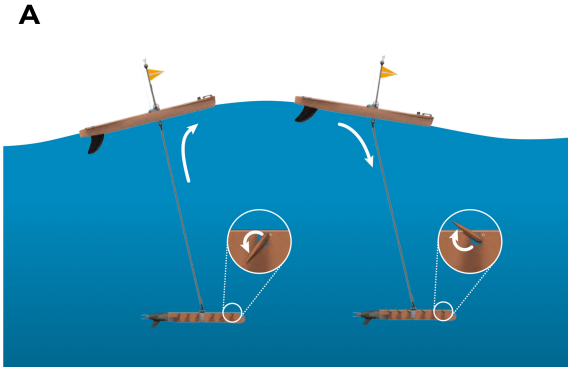
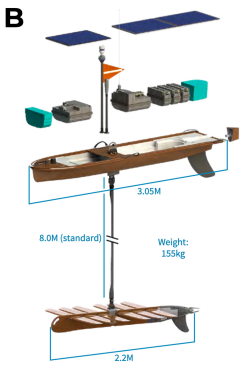
A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2025-04

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined here .		X
Answer is No here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer Yes to any of the above questions.			
If the answer is Yes to Q1 , please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is Yes to Q2 , please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to Q9 is Yes , do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is Yes to Q5 , please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If Yes , provide details of the issue and your plan to solve it]		X

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Student Proposed?	Y/N	N/A
ID	RAV2025-05	
Supervisor	RA Verrinder	
Co-Supervisor(s)	Dr S Nicholson (SOCCO, CSIR)	
Title	Adaptive path planning for Wave Gliders: AIS-based dynamic collision avoidance	
Description	Background Autonomous ocean platforms such as Wave Gliders are frequently used by ocean researchers to observe ocean variables (Patterson et al., 2025). Their unique wave-powered propulsion and long-endurance design enable extended, high-resolution data collection in remote regions, and they are used to measure key variables such as air-sea fluxes of heat and carbon dioxide, as well as ocean temperature, salinity, and dissolved oxygen in dynamic and often poorly observed parts of the global ocean, without the need for ship-based support (Grare et al., 2021; Patterson et al., 2025). However, their full potential is constrained by limited onboard decision-making and path planning capabilities, especially in the harsh and unpredictable conditions of the Southern Ocean and South Africa's coastal waters. South Africa, through the Southern Ocean Carbon–Climate Observatory (SOCCO) (CSIR, 2025), has been a regional leader in deploying Wave Gliders. As these vehicles expand into more challenging and operationally dense marine environments, robust navigation strategies are increasingly required. Autonomous maritime traffic avoidance is critical to reducing risk in high-traffic zones, such as South Africa's coastal shelf regions, where commercial vessel density poses significant operational constraints. Addressing this challenge is essential for extending mission endurance, reducing human oversight, and maximising the scientific return of Wave Glider deployments. Currently, Wave Gliders lack autonomous collision avoidance systems and rely on manual planning and tracking. As a result, their navigation remains reactive and inefficient, and operational risks in coastal zones remain high. The use of AIS (Automatic Identification System) data to dynamically forecast ship trajectories presents a promising solution to improve real-time situational awareness, navigation and safety. This approach could significantly enhance the autonomy and resilience of Wave Gliders operating in complex coastal waters. A  B  Fig. 5. Overview of the Liquid Robotics Wave Glider. (A) A side view schematic illustrating the unique wave driven propulsion of the Wave Glider. (B) An exploded view of a wave glider showing the surface and subsea assemblies as well as the on-board payload. All images taken from Boeing Liquid Robotics (2025). Subject of investigation This project initiates a pilot investigation into AIS-based dynamic collision avoidance for autonomous ocean platforms, with a specific focus on Wave Gliders operating in South Africa's high-traffic coastal zones. The goal is to design and evaluate a system that can autonomously interpret maritime traffic patterns and issue collision avoidance decisions based on predicted vessel movements. The project will explore the use of AIS data streams to detect, classify, and forecast vessel trajectories to improve dynamic path planning for the system. The system will be designed ultimately for real-time integration onto the platform and therefore will need to consider low-latency processing, reliability under communication constraints, and minimal computational overhead. This work will serve as the foundation for a future full-path-planning system that incorporates both dynamic environmental data and maritime safety protocols. By the end of the project, the student will have developed and tested a prototype AIS-aware avoidance system in a simulated operational environment using historical AIS data, contributing to the broader goal of enabling long-endurance, low-risk autonomous operations for climate monitoring in South African waters.	

Deliverables	The key objectives of this project are: (a) To understand the full scope and requirements of maritime traffic-aware navigation for autonomous platforms. (b) To conduct a literature review of AIS-based vessel prediction techniques and collision avoidance algorithms. (c) To acquire and process AIS data to characterise marine traffic in South African coastal waters. (d) To design and implement a prototype system for real-time ship trajectory forecasting using machine learning. (e) To integrate the collision avoidance module into a simulated Wave Glider navigation pipeline. (f) To test and evaluate system performance under representative traffic and environmental conditions. (g) To discuss the system performance, draw conclusions, and make recommendations for future improvements and field deployment. The key deliverables of this project are: (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor through an agreed-upon communication platform, which could include online meetings. (c) A working prototype of a dynamic collision avoidance system using AIS-based ship trajectory forecasting. (d) A simulation framework demonstrating avoidance behaviour under high-traffic conditions. (e) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (f) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (g) A poster summarising the research project report. (h) A seminar to the Department summarising the research project report. (i) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Control systems, Interest in marine science
Extra Information	References G. Rampersadh, M. Finbow, R. Verrinder, and E. Boje. Sea-state interaction based Lagrangian dynamic model of the Liquid Robotics Wave Glider platform. In <i>2024 European Control Conference (ECC)</i>, pages 1119–1124. IEEE, 2024. doi: 10.23919/ECC64448.2024.10591032 L. Grare, N. M. Statom, N. Pizzo, and L. Lenain. Instrumented wave gliders for air-sea interaction and upper ocean research. <i>Frontiers in Marine Science</i>, 8(664728), 2021. doi: 10.3389/fmars.2021.664728 R. G. Patterson, M. F. Cronin, S. Swart, J. Beja, J. M. Edholm, J. McKenna, J. B. Palter, A. Parker, C. I. Addey, W. Boone, P. Bhuyan, J. J. H. Buck, E. F. Burger, J. Burris, L. Camus, B. de Young, M. du Plessis, M. Flanigan, G. R. Foltz, S. T. Gille, L. Grare, J. E. Hansen, L. R. Hole, M. C. Honda, V. Hormann, C. Kohlman, N. Kosaka, C. Kuhn, L. Lenain, L. Looney, A. Marouchos, E. K. McGeorge, C. R. McMahon, S. Mitarai, C. Mordy, A. Nagano, S.-A. Nicholson, S. Nickford, K. M. O'Brien, D. Peddie, L. Ponsoni, V. Ramasco, N. Rozenauers, E. Siddle, C. Stienbarger, A. J. Sutton, N. Tada, J. Thomson, I. Ueki, L. Yu, C. Zhang, and D. Zhang. Uncrewed surface vehicles in the global ocean observing system: a new frontier for observing and monitoring at the air-sea interface. <i>Frontiers in Marine Science</i>, 12(1523585), 2025. doi: 10.3389/fmars.2025.1523585 CSIR. Southern Ocean Carbon Climate Observatory (SOCCO). [ONLINE], 2025. Accessed: 2025-07-11. url: https://socco.org.za/
Broad Research Area	Autonomous Navigation, Marine Robotics
Project suitable for: ME/ECE/EE/ALL?	ME, ECE

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EEE4022S Topic RAV2025-05 GAs

Student Proposed?	Y/N	N/A
ID	RAV2025-05	
Supervisor	RA Verrinder	
Co-Supervisor(s)	Dr S Nicholson (SOCCO, CSIR)	
Title	Adaptive path planning for Wave Gliders: AIS-based dynamic collision avoidance	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem involving real-time collision avoidance in maritime environments using AIS data. It includes ill-posed and under-specified components, such as: - Developing a forecasting algorithm to predict ship trajectories based on noisy and incomplete AIS data. - Integrating trajectory forecasts with autonomous navigation logic in simulation environments. It also presents high-level problems, such as: - Operating in coastal zones with dense and unpredictable ship traffic. - Balancing competing priorities between safety, energy efficiency, and mission coverage. The project demands the handling of complex and incomplete information by: - Processing large volumes of AIS time series data with irregular update intervals. - Designing a decision-making framework under environmental and operational uncertainty.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: - Process and clean AIS data streams to extract trajectory patterns. - Develop and test machine learning models for ship trajectory prediction. - Design and evaluate dynamic path adjustment algorithms in simulation. - Evaluate avoidance strategies under different operational scenarios, including traffic density, route conflicts, and weather variability.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project is computational in nature and will require competencies in time series analysis, simulation, data modelling, and AI-based forecasting. As such, the candidate will demonstrate proficiency in engineering tools and environments used for data analysis, algorithm development, and system integration. Software required may include: - MATLAB/Python for data processing and analysis - MATLAB/Python for modelling and simulation - MATLAB/Python for data validation	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

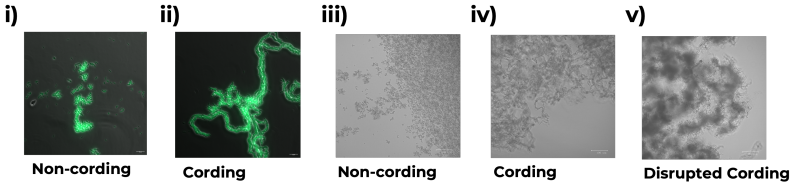
A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2025-05

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined here .		X
Answer is No here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc	X	
Answer the following questions if you answer Yes to any of the above questions.			
If the answer is Yes to Q1 , please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is Yes to Q2 , please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to Q9 is Yes , do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is Yes to Q5 , please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If Yes , provide details of the issue and your plan to solve it]		X

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Student Proposed?	N	N/A
ID	RAV2025-06	
Supervisor	RA Verrinder	
Co-Supervisor(s)	Dr MK Mason (MMRU, IDM)	
Title	Automated quantitative analysis of mycobacterial cording phenotypes	
Description	Quantitative data acquisition and analysis is a key component of microbiological sciences, requiring critical technological skills in sample processing, data generation, acquisition, digitisation, processing and statistical analysis. <i>Mycobacterium tuberculosis</i>, the bacterium that is responsible for causing tuberculosis, has the ability to form cords when cultured in the laboratory under specific growth conditions. Cording is clinically relevant, and has been shown to contribute to the mechanopathology of <i>Mtb</i> (Mishra et al., 2023). Cords have been characterised at high resolution using scanning electron microscopy (Julián et al., 2010) but are also visible microscopically at lower resolutions, can be used as a diagnostic (Caviedes et al., 2000) and are a phenotype of interest (see Figure 6).  Microscopic cording has generally been assessed qualitatively through the comparison of high-resolution images or through the visual categorisation of phenotypes into cording and non-cording (e.g. for the diagnostic MODS assay) (Caviedes et al., 2000). For this project, low magnification images of mycobacterial cells in non-cording aggregates (see Fig. 6, iii), organised cords (see Fig. 6, iv), and disrupted cords (see Fig. 6, v) have been generated as phenotypic readouts of microbiological assays (background strain phenotypes and disruption of cell envelope process). These data form the foundation for a qualitative description of the multicellular organisation of these specific strains and conditions. The aim of this project would be to assess these images and to design and develop an automated image processing, data acquisition, analysis and visualisation solution, which would allow for the quantification and comparison of data for each strain and condition. If more suited, a machine learning approach could be implemented for the classification of distinct phenotypes into specific categories.	
Deliverables	The key objectives of this project are: - to understand the full scope and requirements of the project. - to conduct a literature review of previously published research in this field and critically evaluate current technologies and research that applies to this project. - to familiarise self with image dataset and curate dataset in a manner appropriate to automated data processing (file re-organisation, re-naming, categorisation/labelling). - to explore image analysis techniques that allow for the quantification of complex features and textures. - to pre-process image data to allow for uniformity background (to accommodate acquisition artifacts and allow standardisation within and among experiments) - to process image data to allow bacterial aggregate or cord feature recognition and data acquisition. - to quantify cording phenotypes / features to allow for analysis (comparisons among samples) - OPTIONAL: explore and implement a machine learning approach to classify images in terms of their cording phenotype. - test and evaluate the system performance and compare to previously implemented solutions. - discuss the performance of the system, draw conclusions and make recommendations for future improvements. The key deliverables of this project are: - A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project) - Weekly reporting of project progress and plans to the supervisor(s) through an agreed upon communication platform, which could include online meetings - An image pre-processing, processing and data analysis pipeline that is suited for the quantification of cording phenotypes for the images provided. - A research project report, following the Department of Electrical Engineering Final Year Project report guidelines - Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs) - A poster summarising the research project report - A seminar to the Department summarising the research project report - Attendance and presentation at the annual Department Open Day.	

Skills/Requirements	Python, Mathematical skills, Data analysis and visualisation, Statistical competencies, Interest in biomedical engineering
Extra Information	References L. Caviedes, T. S. Lee, R. H. Gilman, P. Sheen, E. Spellman, E. H. Lee, D. E. Berg, and S. Montenegro-James. Rapid, efficient detection and drug susceptibility testing of <i>Mycobacterium tuberculosis</i> in sputum by microscopic observation of broth cultures. <i>Journal of Clinical Microbiology</i>, 38(3):1203–1208, Mar 2000. doi: 10.1128/JCM.38.3.1203-1208.2000 E. Julián, M. Roldán, A. Sánchez-Chardi, O. Astola, G. Agustí, and M. Luquin. Microscopic cords, a virulence-related characteristic of <i>Mycobacterium tuberculosis</i>, are also present in nonpathogenic mycobacteria. <i>Journal of Bacteriology</i>, 192(7):1751–1760, Apr 2010. doi: 10.1128/JB.01485-09 R. Mishra, M. Hannebelle, V. P. Patil, A. Dubois, C. Garcia-Mouton, G. M. Kirsch, M. Jan, K. Sharma, N. Guex, J. Sordet-Dessimoz, J. Perez-Gil, M. Prakash, G. W. Knott, N. Dhar, J. D. McKinney, and V. V. Thacker. Mechanopathology of biofilm-like <i>Mycobacterium tuberculosis</i> cords. <i>Cell</i>, 186(23):5135–5150.e28, Nov 2023. doi: 10.1016/j.cell.2023.09.016 Potential for Publication The dataset provided by Dr Mason comprises a set of low magnification images of <i>Mycobacterium tuberculosis</i> CRISPRi hypomorph strains that have been grown in liquid culture. These images represent a novel dataset of bacterial strains that are being characterised for publication. The demonstration of a tractable and complete implementation (a full, user friendly working solution) to automatically quantify (or categorise) phenotypes has the potential to complement (and contribute to) data that are planned for publication. Work performed in this project that is suited for inclusion in this manuscript will result in co-authorship for this publication. Students who are interested in contributing to publishable research and who would be interested in being involved in the publication process are encouraged to apply for this project.
Broad Research Area	Image processing and analysis, Machine Learning, Data analysis and visualisation, Microbiology, Tuberculosis.
Project suitable for: ME/ECE/EE/ALL?	ECE

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EEE4022S Topic RAV2025-06 GAs

Student Proposed?	N	N/A
ID	RAV2025-06	
Supervisor	RA Verrinder	
Co-Supervisor(s)	Dr MK Mason (MMRU, IDM)	
Title	Automated quantitative analysis of mycobacterial cording phenotypes	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	The images provided represent a complex problem for preprocessing as they: - comprise complex imaging data. - are not standardised (contain large variations and artefacts). The data generated present a complex problem for processing as they: - Reflect data of a biological origin that may be unfamiliar and require the candidate to learn about a new field (new concepts). - Must be examined in several ways to reach the best solution for how to segment highly variable features. The data analysis component will also be complex and require novel solutions as: - The quantification of cords is an open-ended problem. - Statistical analysis of data is desirable. - Image data that have been generated are highly complex and variable and represent a novel phenotype and various analysis approaches must be explored.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: - Explore segmentation, feature identification approaches to determine if/how data can be quantified. - Explore machine learning approaches to determine, if these should be utilised, to categorise data (strains and conditions) into meaningful groups.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This topic, which provides an image dataset for processing and analysis, will be computational in nature and will require competencies in file handling, image processing, data handling, and data visualisation. Due to the complex nature of the images provided, it is suggested that a Machine Learning approach could be explored for a categorisation approach. As such, the candidate will demonstrate competency in the use of computing and software and must select the appropriate data handling processes and software tools to accomplish their objectives. Software required may include: - Python/MATLAB - ImageJ/FIJI	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2025-06

Question	Question Text	Yes	No
Q1	Does this project involve data collection		X
Q2	Does this project involve utilizing a third-party data set	X	
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?	X	
Q4	Does it exceed the minimum risk defined here ?		X
Answer is No here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer Yes to any of the above questions.			
If the answer is Yes to Q1 , please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is Yes to Q2 , please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)	X	
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to Q9 is Yes , do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is Yes to Q5 , please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If Yes , provide details of the issue and your plan to solve it]		X

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Student Proposed?	Y/N	N/A
ID	RAV2025-07	
Supervisor	RA Verrinder	
Co-Supervisor(s)	RM Bernartzeder; R Bernartzeder	
Title	Remote weight monitoring system for top hives in the Zambian Miombo woodlands	
Description	Background The Miombo woodlands of Zambia provide an essential ecological and economic foundation for many rural communities. Forest beekeeping, a long-established tradition among these communities, offers a significant and sustainable source of income. Furthermore, the indigenous African honey bee, <i>Apis mellifera</i>, plays a crucial role in the forest ecosystem as the primary pollinator of numerous native tree species within this biome. Traditional beekeeping practices involve the construction of simple hives and suspending them high in trees. These top hives are designed to attract wild swarms and are strategically placed to protect against predators and support optimal honey production. While effective in preserving the hives and promoting bee health, this method presents substantial logistical challenges for beekeepers (Lowore et al., 2018; Lowore, 2020; Meaton et al., 2020). Harvesting honey from these elevated hives requires strenuous physical effort and extensive travel, often taking several days. As a result, beekeepers may only visit their hives once or twice a year. Critically, due to the absence of information regarding hive condition and honey yield, such trips often result in small harvests. These inefficiencies not only waste time and energy but also detract from other essential agricultural activities vital to food security and the economic well-being of rural households.  Fig. 7. Beekeeping in the Zambian Miombo woodlands. (A) A top-hive hung in a tree; (B) Beekeepers retrieving the hive to harvest honey; (C) A honey comb from one of the hives. (Photos credit: Robert Bernartzeder, 2024) Subject of investigation This project addresses the need for a practical solution that enables remote monitoring of honey production in forest hives. A real-time hive weight and environmental monitoring system would allow beekeepers to make informed decisions about when to harvest, thereby eliminating unnecessary trips. The primary project objective is to design and prototype an electronic scale capable of accurately measuring the weight of elevated beehives and transmitting this information remotely to farmers. The system must be accurate, durable enough for the harsh woodland environment, energy-efficient to ensure operational longevity, and user-friendly to accommodate users with varying levels of technical experience. This project aims to reduce labour burdens, improve honey production, and allow beekeepers to better allocate their time between multiple livelihood activities. By integrating electronic engineering solutions into rural beekeeping, the project will not only support sustainable agriculture, but also empower communities through technology-driven resilience (Frazier et al., 2024; Lowore et al., 2018; Lowore, 2020; Enhanced Integrated Framework (EIF), 2022). This project is a collaboration between the UCT African Robotics Unit and the Zambian Meadery.	

Deliverables	The key objectives of this project are: (a) To understand the full scope and requirements of the project. (b) To conduct a literature review of previously published research in this field and critically evaluate current technologies and research that applies to this project. (c) To design and develop an electronic scale and environmental sensing prototype for remote measurement. (d) To integrate a reliable data transmission system using technologies such as Cellular modems/Iridium SBD. (e) To test and evaluate the system performance in real-world conditions. (f) To discuss the performance of the system, draw conclusions, and make recommendations for future improvements. The key deliverables of this project are: (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor through an agreed-upon communication platform, which could include online meetings. (c) A fully functional prototype for remote measurement. (d) A data transmission system that reliably transmits hive data to beekeepers. (e) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (f) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (g) A poster summarising the research project report. (h) A seminar to the Department summarising the research project report. (i) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	C programming, Embedded systems, Interest in sustainable agriculture
Extra Information	References M. Frazier, E. Muli, and H. Patch. Ecology and management of African honey bees (<i>Apis mellifera</i> L.). <i>Annual Review of Entomology</i>, 69:439–453, Jan 2024. ISSN 0066-4170. doi: 10.1146/annurev-ento-020823-095359 J. Lowore, J. Meaton, and A. Wood. African forest honey: an overlooked NTFP with potential to support livelihoods and forests. <i>Environmental Management</i>, 62(1):15–28, 2018. doi: 10.1007/s00267-018-1015-8 J. Lowore. Understanding the livelihood implications of reliable honey trade in the Miombo woodlands in Zambia. <i>Frontiers in Forests and Global Change</i>, 3, 2020. doi: 10.3389/ffgc.2020.00028 J. Meaton, J. Lowore, and A. Wood. Assessing value chain interventions in Zambian and Ethiopian forest beekeeping systems. <i>Business Strategy and Development</i>, 4(2):159–169, Aug 2020. ISSN 2572-3170. doi: 10.1002/bsd2.136 UNESCO. Creating a buzz in Zambia: beekeeping for sustainable development. [ONLINE], 2021. Accessed: 2024-06-25. url: https://www.unesco.org/en/articles/creating-buzz-zambia-beekeeping-sustainable-development Enhanced Integrated Framework (EIF). From hunting to harvesting: Zambia's eco-friendly approach to beekeeping. [ONLINE], 2022. Accessed: 2024-06-25. url: https://trade4devnews.enhancedif.org/en/impact-story/hunting-harvesting-zambias-eco-friendly-approach-beekeeping
Broad Research Area	Sustainable agriculture, Beekeeping, Embedded systems, IoT
Project suitable for: ME/ECE/EE/ALL?	All

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EEE4022S Topic RAV2025-07 GAs

Student Proposed?	Y/N	N/A
ID	RAV2025-07	
Supervisor	RA Verrinder	
Co-Supervisor(s)	RM Bernartzeder; R Bernartzeder	
Title	Remote weight monitoring system for top hives in the Zambian Miombo woodlands	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the development of an electronic scale and remote monitoring system for beekeeping. It involves ill-posed and under-specified problems, such as the: - refinement of remote monitoring technology for hive weight measurement - development of data transmission algorithms using cellular and/or Iridium SBD, which can be run on an embedded system. It also presents high-level problems, such as: - Understanding weight measurement in varying environmental conditions. - Calibrating sensors and ensuring accurate weight measurement under different hive designs. - Handling harsh environmental factors and power constraints. - Solutions for integrating data transmission technology into the electronic scale system, which is not trivial. The project also necessitates handling complex and incomplete information by: - Synthesising information from various sources such as beekeeping practices, sensor specifications, and data transmission requirements. - Balancing technical challenges, engineering design, and stakeholder interests.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: - Design and prototype the electronic scale for hive weight measurement. - Develop and test the data transmission system using cellular or Iridium SBD. - Validate the accuracy and reliability of the weight measurements in real-world conditions. - Analyze the transmitted data to ensure it meets the requirements for remote monitoring.	
	This project, which involves the development of an electronic scale and remote monitoring system, will require competencies in embedded systems, data transmission, and data analysis. As such the candidate will demonstrate competency in the use of computing and software and must select the appropriate data handling processes and software tools to accomplish their objectives. Software and tools required may include: - C programming for embedded systems. - MATLAB/Python for data processing and analysis - MATLAB/Python for modelling and simulation - MATLAB/Python for data validation	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
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- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

Ethics clearance questionnaire: RAV2025-07

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined here .		X
Answer is No here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc	X	
Answer the following questions if you answer Yes to any of the above questions.			
If the answer is Yes to Q1 , please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is Yes to Q2 , please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to Q9 is Yes , do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is Yes to Q5 , please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If Yes , provide details of the issue and your plan to solve it]		X

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